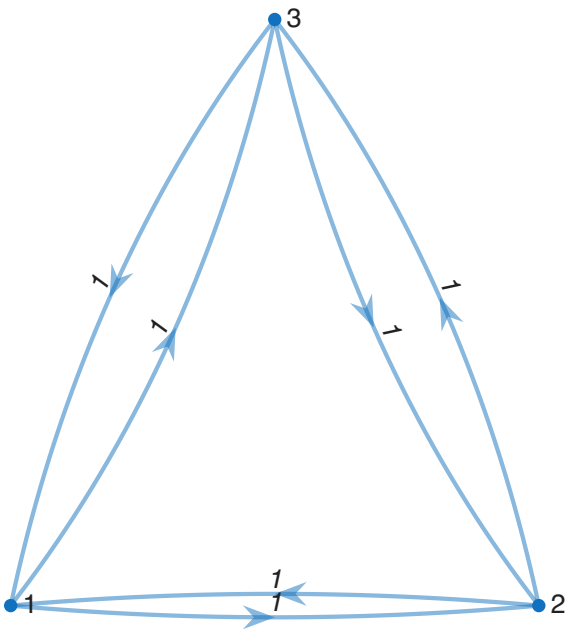
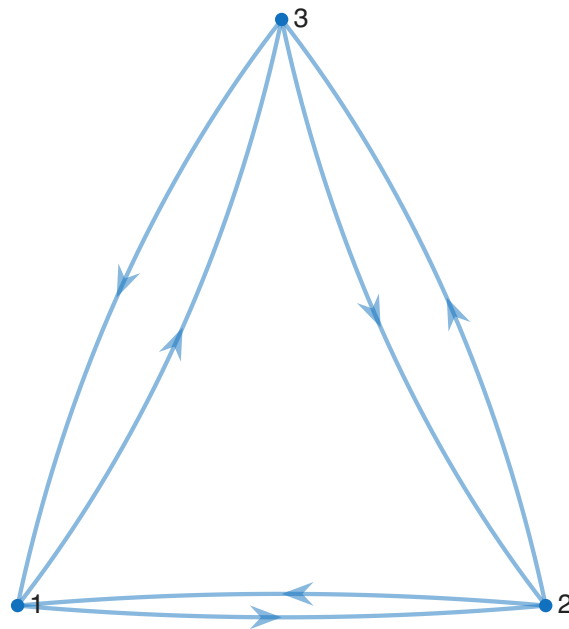


(a) Same circulation C 

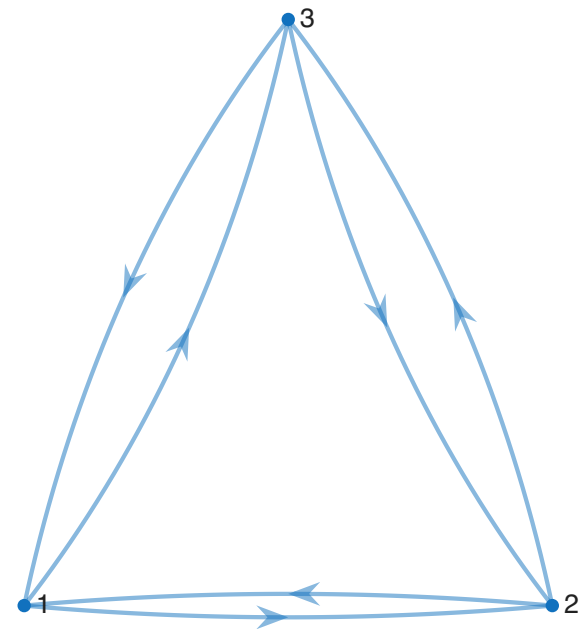
Unit bidirected triangle
 $V_C = \sum_e c_e = 6$

(b) Three two-cycle representation



$$\begin{aligned}\lambda_{(1,2)} &= \lambda_{(1,3)} = \lambda_{(2,3)} = 1 \\ T_C &= 1 + 1 + 1 = 3 \\ V_C &= 3 \times 2 = 6\end{aligned}$$

(c) Two three-cycle representation



$$\begin{aligned}\lambda_{(1,2,3)} &= \lambda_{(1,3,2)} = 1 \\ T_C &= 1 + 1 = 2 \\ V_C &= 2 \times 3 = 6\end{aligned}$$

The same circulation has invariant edge-volume $V_C=6$, but different completed throughput T_C under different cycle representations.